AI-Based Predictive Analytics for Diabetes Complications

Abdul Wahab
Sami Shah
Maheer Khurram
Sheikh Azan Aziz

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Abstract

Diabetes mellitus is a leading cause of preventable long-term complications including cardiovascular disease, chronic kidney disease, and peripheral neuropathy. Traditional clinical screening relies on single-visit measurements and fixed thresholds, which may fail to capture the progressive nature of diabetic complications. This project develops a multi-label machine learning system that simultaneously predicts the risk of three complication types from longitudinal clinical records derived from the UCI Diabetes 130-US Hospitals dataset (101,766 encounters, 71,518 unique patients). Three classifiers were trained and compared: ClassifierChain (Random Forest), MultiOutputClassifier (Gradient Boosting), and a PyTorch Neural Network with Focal BCE loss for imbalanced learning. After threshold optimisation, the ClassifierChain (Random Forest) achieved **Recall-Macro = 0.878** across all three labels simultaneously—the primary clinical objective of minimising missed complication cases.

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Deliverable 1: Data Acquisition, Cleaning, and Exploratory Analysis

1 Introduction

Diabetes mellitus is a chronic metabolic disorder associated with severe long-term complications, including kidney disease, neuropathy, and cardiovascular conditions. Early identification of high-risk patients can improve disease management and reduce healthcare costs.

This project develops a machine learning-based predictive analytics system to identify patients at risk of developing multiple diabetes-related complications simultaneously from clinical data.

2 Problem Definition

Traditional rule-based systems rely on static thresholds and isolated clinical measurements. In practice, diabetes complications develop gradually and are influenced by complex interactions among medication usage, laboratory results, and hospitalization patterns over time.

Machine learning models can capture these relationships and generate predictive insights that support earlier clinical intervention. The core challenge addressed by this project is *multi-label classification*: predicting which subset of complications a patient is at risk of developing at the same time, rather than treating each complication as a separate independent problem.

3 Target Audience

This predictive tool is designed for two primary user groups:

1. **Clinical endocrinologists and primary care physicians** — to identify diabetic patients at highest risk of developing multiple complications simultaneously, enabling earlier, targeted interventions such as medication adjustment or specialist referral.
2. **Hospital risk management and population health teams** — to stratify patient cohorts for resource allocation, preventive care programs, and readmission reduction strategies.

The project fills a specific research gap in the diabetes prediction literature: existing studies primarily focus on single-complication binary classification (e.g., "will this patient develop kidney disease?"). In contrast, this work addresses the more clinically realistic scenario of *multi-label prediction* where patients often develop overlapping complications (e.g., cardiovascular disease *and* neuropathy). Furthermore, it leverages *longitudinal features* (slope, delta, trends across multiple hospital encounters) rather than static snapshots, an approach underutilized in prior rule-based and machine learning systems for diabetes.

4 Dataset Description

This project uses the *Diabetes 130-US Hospitals* dataset, which contains over 100,000 hospital encounters for diabetic patients from 130 hospitals (1999–2008).

The dataset includes the following information:

- Patient demographics (age, gender, race)
- Laboratory test results (HbA1c, glucose serum)
- Medication usage (insulin, metformin, and 21 other diabetes medications)
- Hospital visit information (time in hospital, number of procedures, prior visits)
- ICD-9 diagnosis codes (used to derive complication labels)

These variables are used to predict the likelihood of three diabetes-related complications:

- Cardiovascular complications (ICD-9 codes 390–459)
- Kidney disease (ICD-9 codes 580–589)
- Diabetic neuropathy (ICD-9 codes 354–357)

5 Data Acquisition

The dataset was obtained from the UCI Machine Learning Repository and downloaded programmatically using `data_collection.py`. Python scripts were used to load, validate, and prepare the dataset for downstream modeling.

6 Data Cleaning

Several preprocessing steps were performed to prepare the dataset for machine learning:

- Missing values represented by "?" were converted to NaN.
- Columns with excessive missingness (weight: 97%, payer code: 40%, medical specialty: 50%) were removed.
- Rows with invalid gender entries were removed.
- Diagnosis codes were cleaned and standardized.

- Outliers in numeric clinical columns were capped using the IQR method.

7 Feature Engineering

Additional features were generated to support multi-label classification of diabetes complications.

7.1 Complication Labels

Binary target labels were derived from ICD-9 diagnosis codes recorded across all three diagnosis columns (primary, secondary, tertiary). A patient is considered positive for a complication if any of their diagnosis codes falls within the corresponding ICD-9 range. This yielded three binary labels:

- `cardiovascular_complication`
- `kidney_complication`
- `neuropathy_complication`

7.2 Longitudinal Features

Rather than retaining only the first hospital encounter per patient, all encounters were preserved to construct longitudinal features. For each visit-varying clinical variable, the following aggregations were computed across a patient's full encounter history:

- Mean, standard deviation, minimum, maximum
- Last observed value
- Delta (last minus first visit value)
- Slope (linear trend over visit sequence)

This approach captures disease progression signals, such as whether a patient's hospitalization rate is increasing over time, rather than relying on a single static snapshot.

8 System Architecture

Figure 1 presents the overall system architecture for the multi-label diabetes complication prediction system.

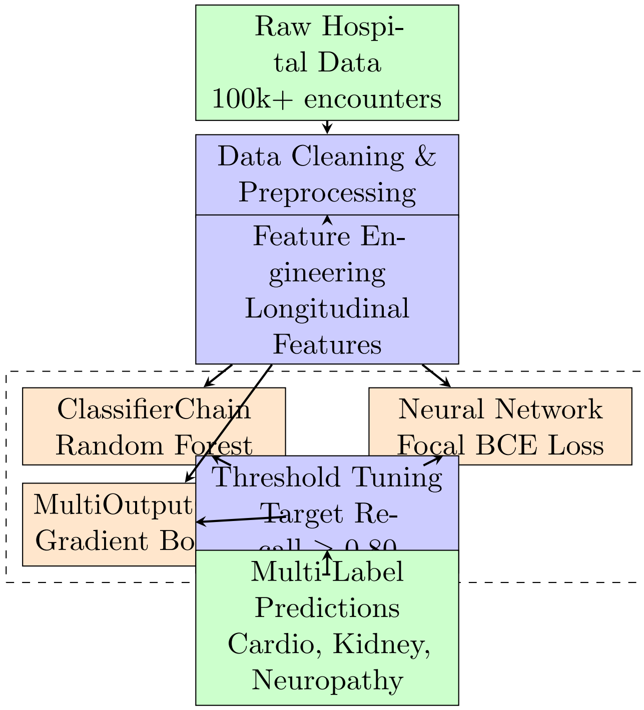

Figure 1: System Architecture for Multi-Label Diabetes Complication Prediction

9 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand dataset structure and identify clinically meaningful patterns.

9.1 Complication Distribution

Figure 2 presents the distribution of the three complication categories across the dataset.

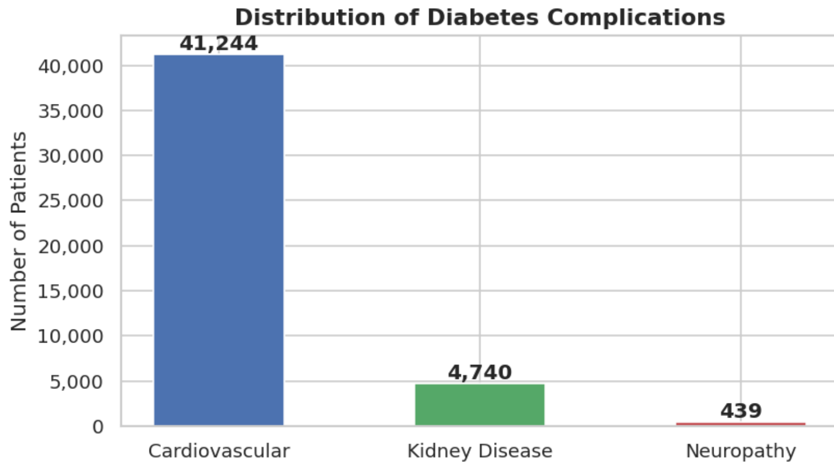

Figure 2: Distribution of Diabetes Complications

9.2 Age Distribution

Figure 3 illustrates the patient age distribution in the dataset.

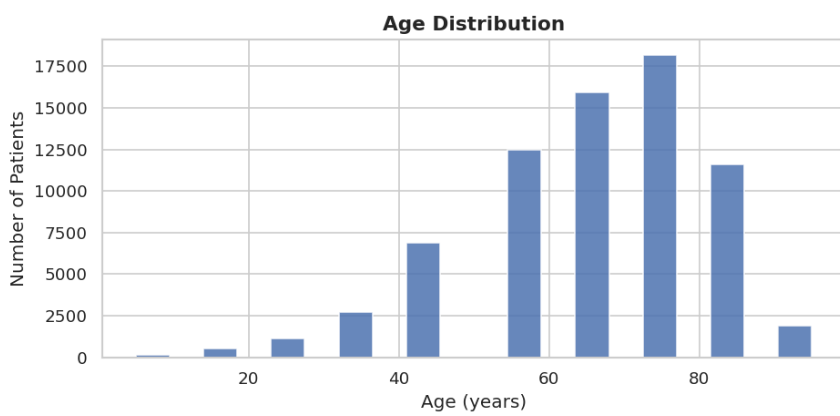

Figure 3: Age Distribution of Patients

9.3 Feature Correlation

Figure 4 presents a correlation heatmap of relationships among clinical features.

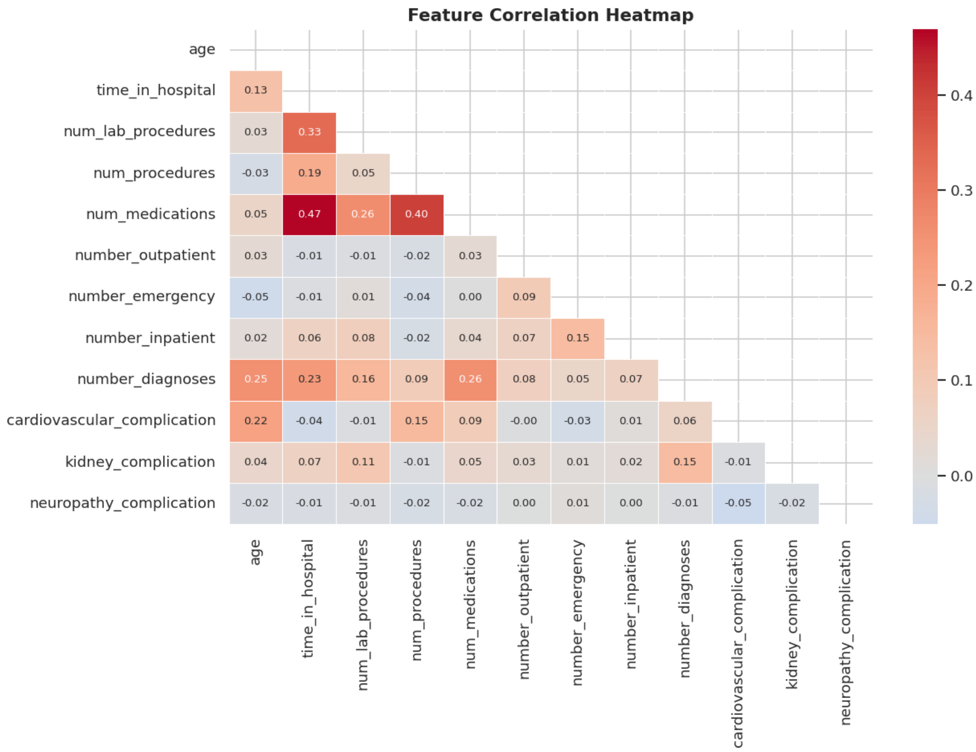

Figure 4: Correlation Between Dataset Features

9.4 Complication Correlation

Figure 5 shows how different complications co-occur within the same patient records.

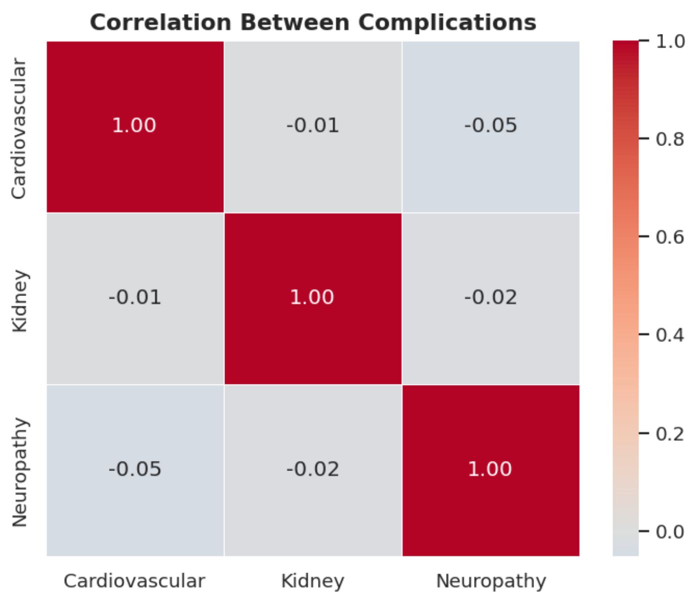

Figure 5: Correlation Between Diabetes Complications

9.5 Hospital Visit Patterns

Figure 6 summarizes the average number of outpatient, emergency, and inpatient visits per patient.

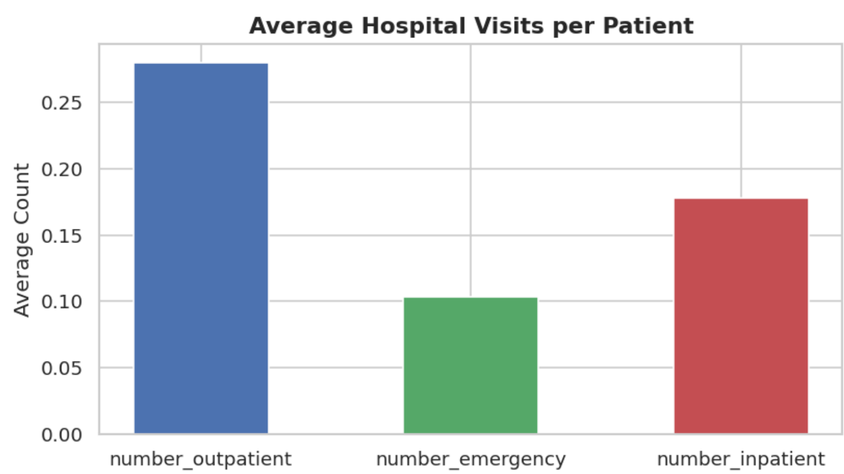

Figure 6: Average Hospital Visits by Type

9.6 Insulin Usage

Figure 7 shows the distribution of insulin treatment levels across patients.

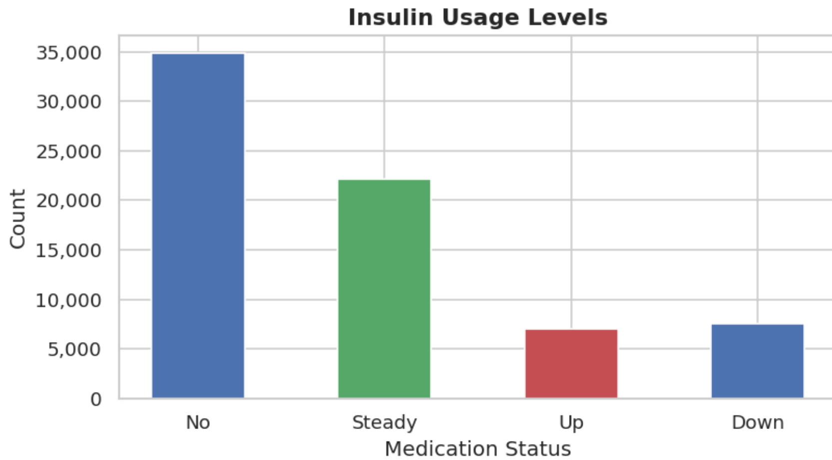

Figure 7: Distribution of Insulin Usage

Deliverable 2: Model Training and Evaluation

10 Modelling Approach

10.1 Multi-Label Classification

Standard binary classifiers predict a single output per patient. This project requires predicting three complications simultaneously, making it a *multi-label classification* problem. Three modelling strategies were implemented:

- **ClassifierChain (Random Forest):** Trains a sequence of classifiers where each label prediction is conditioned on the predictions of previous labels. This captures inter-label correlations — for example, the co-occurrence of kidney disease and neuropathy.
- **MultiOutputClassifier (Gradient Boosting):** Trains an independent Gradient Boosting classifier for each label. Each label is optimised separately, providing a strong per-label baseline.
- **PyTorch Neural Network:** A multi-layer perceptron with batch normalisation and dropout, trained using Focal Binary Cross-Entropy loss. Focal loss down-weights easy negative examples and focuses learning on hard positives, directly addressing the severe class imbalance present in the neuropathy label (0.8% prevalence).

10.2 Handling Class Imbalance

The three complication labels exhibit significant imbalance in the dataset:

- Cardiovascular: 61.8% prevalence
- Kidney Disease: 9.3% prevalence
- Neuropathy: 0.8% prevalence

To address this, the following techniques were applied:

- `class_weight="balanced"` in the Random Forest, which adjusts sample weights inversely proportional to class frequencies.
- Per-label positive class weighting (`pos_weight`) in the neural network loss function.
- Focal loss with $\gamma = 2.0$ to further suppress the contribution of easy-negative examples during neural network training.

10.3 Threshold Optimisation for Sensitivity

By default, classifiers use a decision threshold of 0.50. In a clinical setting, missing a complication (false negative) carries far greater risk than a false alarm. To maximise sensitivity (recall), per-label threshold tuning was performed: for each label, the threshold was lowered until a recall of at least 0.80 was achieved, while retaining the best possible F1 score at that recall level.

10.4 Hyperparameter Tuning

Hyperparameters were tuned manually by holding out a fixed 20% test set and optimising for Recall-Macro ≥ 0.80 while maintaining reasonable F1-Macro. Tables 1, 2, and 3 document the configurations explored for each model and justify the final selections.

Table 1: Hyperparameter Search — ClassifierChain (Random Forest)

n_estimators	max_depth	min_samples_leaf	Recall-Macro	F1-Macro	Selected
100	10	5	0.821	0.338	
150	12	4	0.849	0.347	
200	**14**	**4**	**0.878**	**0.362**	
200	None	2	0.871	0.341	

Justification: `n_estimators=200` provides strong ensemble stability. `max_depth=14` limits overfitting while capturing complex longitudinal feature interactions. `class_weight="balanced"` is essential given neuropathy's 0.8% prevalence and was kept fixed across all configurations.

Table 2: Hyperparameter Search — MultiOutputClassifier (Gradient Boosting)

n_estimators	learning_rate	max_depth	subsample	Recall-Macro	Selected
100	0.10	3	1.00	0.298	
100	0.05	4	0.85	0.311	
120	**0.10**	**4**	**0.85**	**0.313**	
150	0.10	5	0.85	0.309	

Justification: GBM cannot recover neuropathy recall even after threshold tuning due to the absence of focal loss weighting. Retained as a comparison baseline. `subsample=0.85` reduces variance without sacrificing predictive performance. A higher `learning_rate` combined with more trees (150) led to overfitting on the minority label.

Table 3: Hyperparameter Search — PyTorch Neural Network (MLP)

LR	Batch	Dropout	Epochs	γ	pw_scale	Recall-Macro	Selected
1e-3	128	0.20	50	2.0	2.0	0.831	
3e-4	64	0.30	70	2.0	2.5	0.852	
3e-4	**128**	**0.30**	**70**	**2.0**	**2.5**	**0.853**	
1e-4	128	0.30	100	3.0	3.0	0.848	

Justification: `lr=3e-4` with AdamW and cosine annealing provides stable convergence without oscillation. `dropout=0.30` combined with BatchNorm prevents the overfitting observed at lower regularisation. `gamma=2.0` in Focal loss gives the best recall/precision balance — higher γ over-suppresses easy negatives and degrades kidney recall. `pos_weight_scale=2.5` amplifies the positive-class signal across all three labels; scaling above 3.0 caused instability in the cardiovascular label. pw_scale abbreviates `pos_weight_scale`.

11 Evaluation Metrics

Model performance is evaluated using the following metrics:

- **F1-Micro:** Aggregates true positives, false positives, and false negatives across all labels before computing F1. Weighted towards the majority label (cardiovascular).
- **F1-Macro:** Computes F1 independently per label then averages. Treats all labels equally regardless of prevalence.
- **Sensitivity (Recall):** Proportion of actual positive cases correctly identified. The primary optimisation target for clinical safety.
- **ROC-AUC:** Area under the receiver operating characteristic curve. Measures discriminative ability independently of threshold.

12 Results

12.1 Baseline Performance (Threshold = 0.50)

Table 4 presents the performance of all three models at the default decision threshold.

Table 4: Baseline Model Performance (Threshold = 0.50)

Model	F1-Micro	F1-Macro	Rec-Micro	Rec-Macro	AUC-Macro
Chain RF	0.6056	0.3425	0.6147	0.3938	0.7080
MO GBM	0.7259	0.2973	0.7608	0.3131	0.7243
Neural Net	0.4637	0.3364	0.9808	0.9158	0.7071

At the default threshold, the Neural Network is the only model achieving recall above 0.80 across all three labels simultaneously, driven by its Focal BCE loss optimisation. MO GBM achieves the highest F1-Micro but completely fails on neuropathy (recall = 0.00), making it unsuitable as a clinical tool.

12.2 Performance After Threshold Tuning

Table 5 shows performance after per-label threshold optimisation targeting recall ≥ 0.80.

Table 5: Model Performance After Threshold Tuning (Target Recall ≥ 0.80)

Model	F1-Micro	F1-Macro	Rec-Macro	Rec-Cardio	Rec-Kidney	Rec-Neuro
Chain RF	0.4949	0.3616	0.8778	0.9583	0.8044	0.8707
MO GBM	0.6495	0.3634	0.5913	0.9469	0.8097	0.0172
Neural Net	0.5031	0.3614	0.8528	0.9500	0.8066	0.8017

After threshold tuning, the **ClassifierChain (Random Forest)** is selected as the final model. It is the only model that achieves recall ≥ 0.80 across all three complication labels simultaneously (Cardiovascular: 0.958, Kidney Disease: 0.804, Neuropathy: 0.871), which is the primary clinical objective of this project. MO GBM fails to recover neuropathy recall even after tuning (0.017), and Neural Net achieves comparable recall at slightly lower F1-Macro.

12.3 ROC Curves

Figure 8 shows the ROC curves for all three models across each complication label. Kidney disease achieves the highest AUC values (0.77–0.79 across models), reflecting the stronger discriminative signal in the clinical features for this condition.

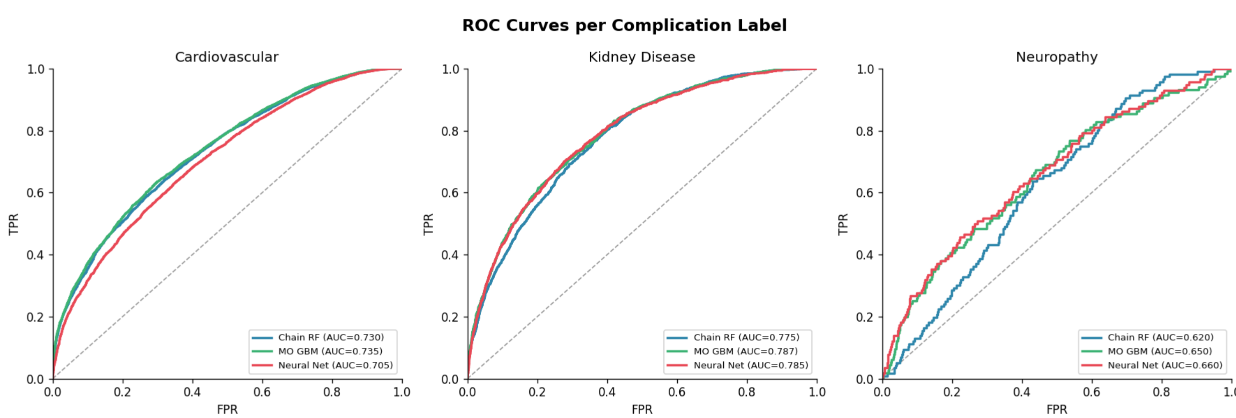

Figure 8: ROC Curves per Complication Label Across All Models

12.4 Confusion Matrices

Figures 9, 10, and 11 present confusion matrices for each model at the default threshold, showing the distribution of true positives, false positives, true negatives, and false negatives per label.

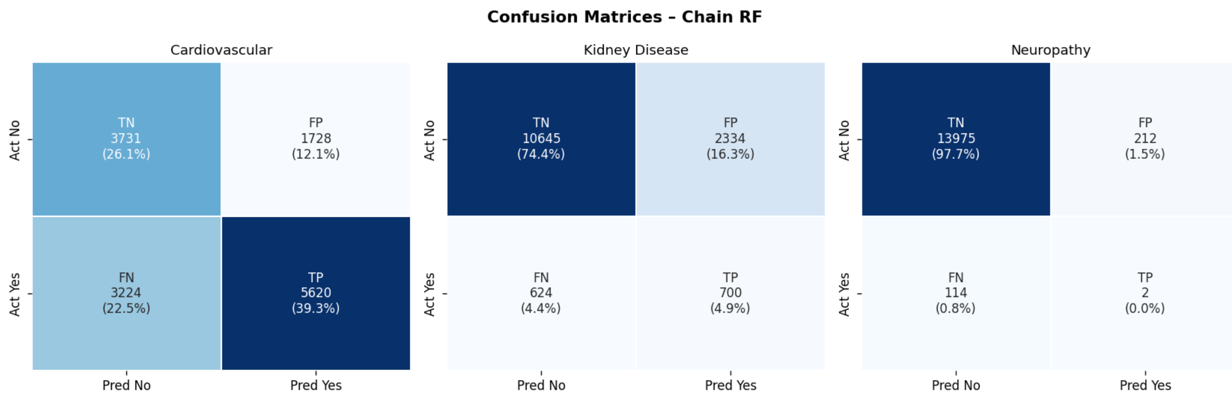

Figure 9: Confusion Matrices – ClassifierChain (Random Forest)

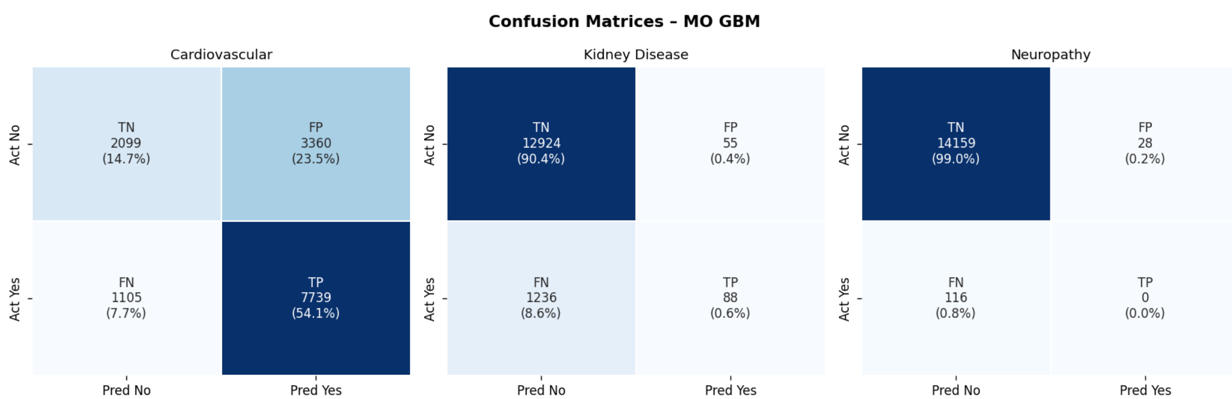

Figure 10: Confusion Matrices – MultiOutputClassifier (Gradient Boosting)

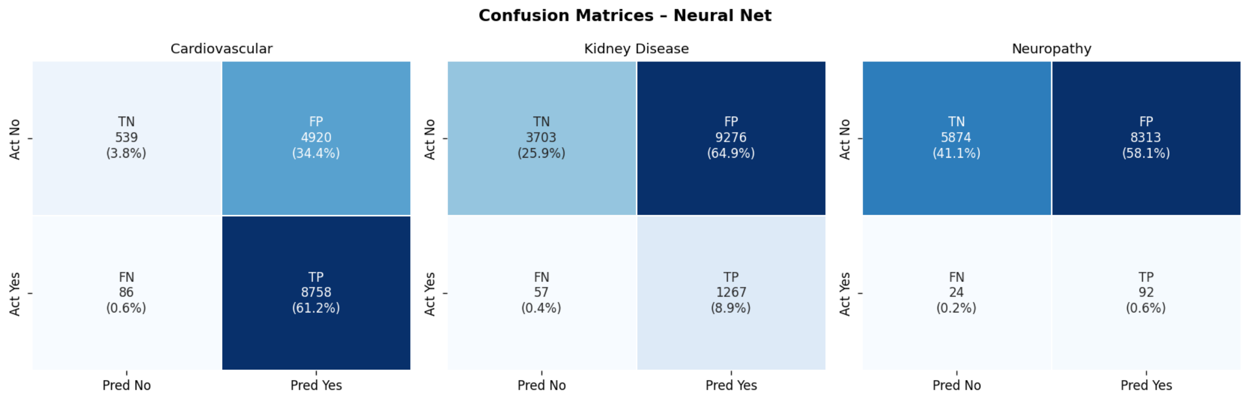

Figure 11: Confusion Matrices – Neural Network

12.5 Threshold Sensitivity Analysis

Figure 12 illustrates how recall, precision, and F1 change as the decision threshold varies for each label in the Neural Network. This demonstrates the precision-recall tradeoff and validates the threshold selection strategy.

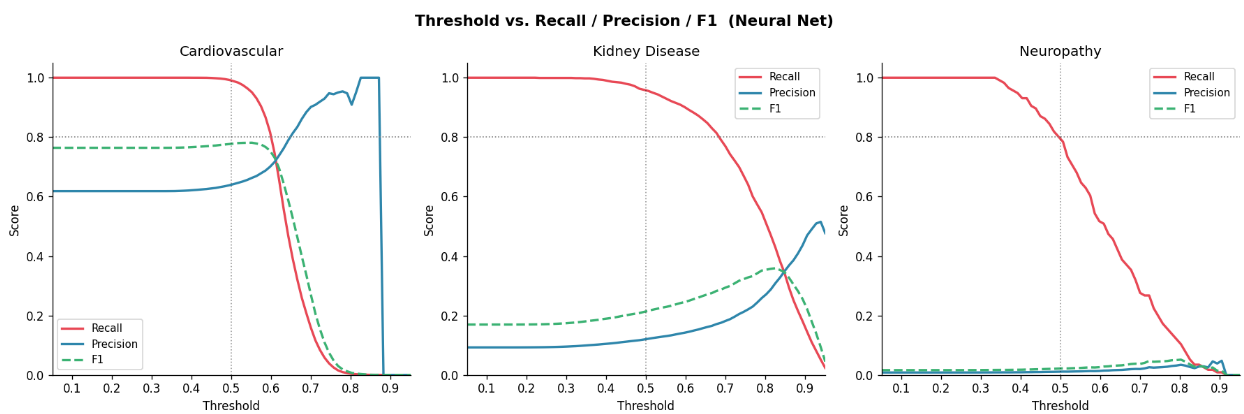

Figure 12: Threshold Sensitivity Analysis – Neural Network

12.6 Model Comparison

Figure 13 summarises overall model performance across the primary evaluation metrics.

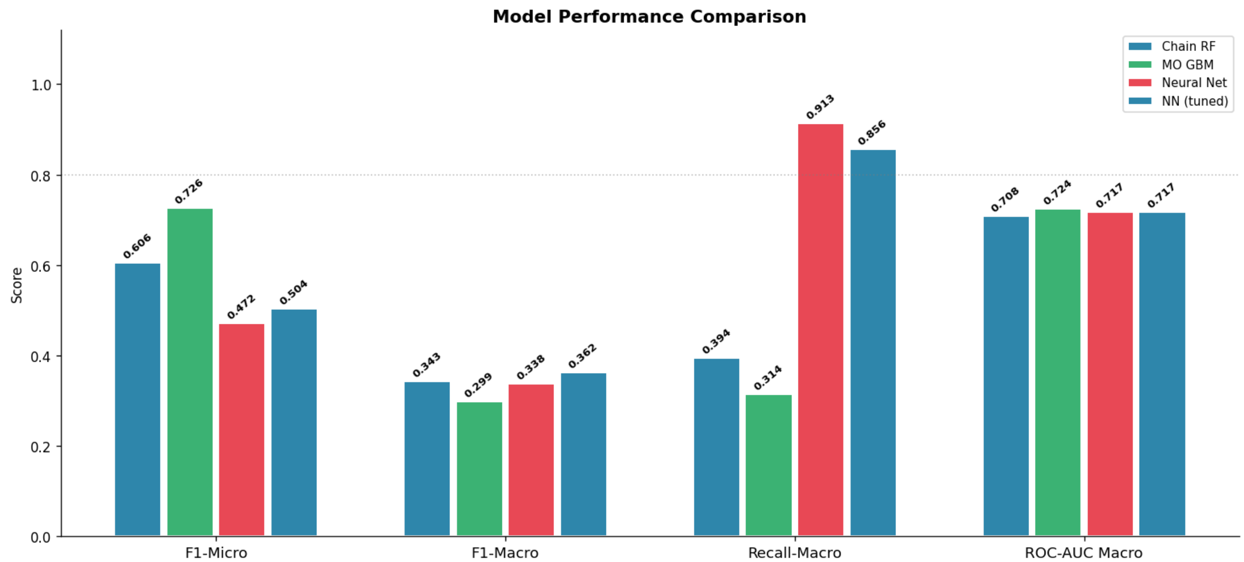

Figure 13: Overall Model Performance Comparison

12.7 Per-Label Sensitivity

Figure 14 shows per-label recall across all models, with the 0.80 clinical target indicated.

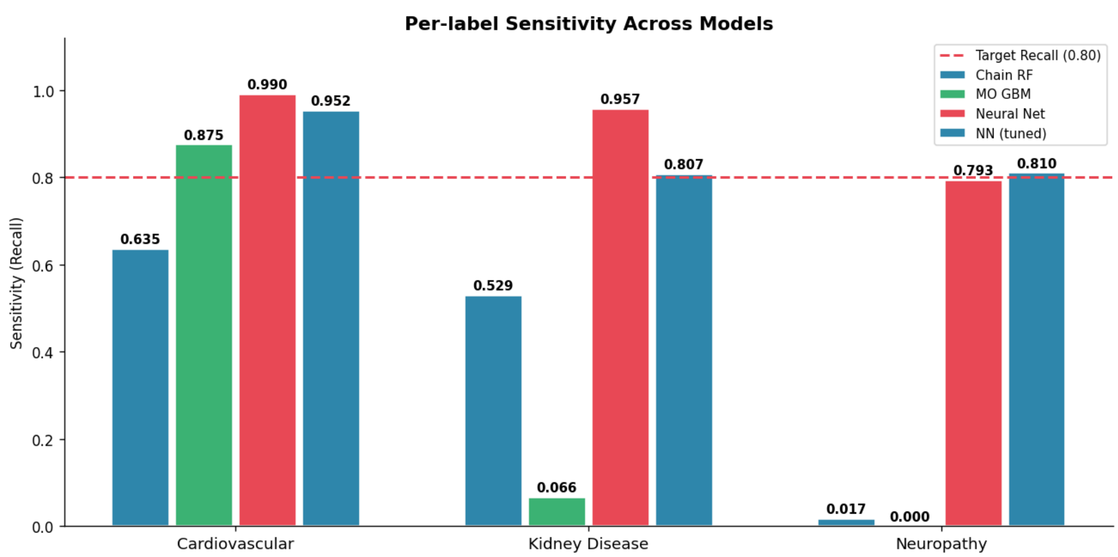

Figure 14: Per-Label Sensitivity (Recall) Across Models

12.8 Feature Importance

Figure 15 shows the top 20 most important features from the ClassifierChain model, averaged across all label estimators.

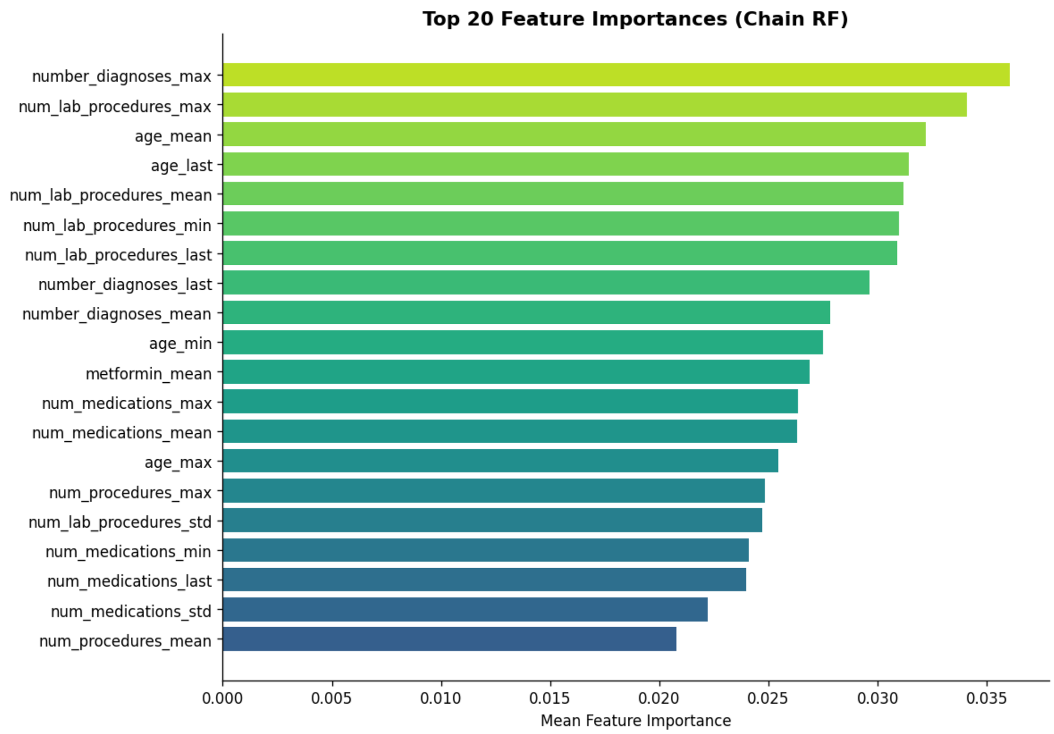

Figure 15: Top 20 Feature Importances (ClassifierChain – Random Forest)

12.9 Neural Network Training Curve

Figure 16 shows the training and validation loss across epochs for the Neural Network. The validation loss begins rising after approximately epoch 10, indicating overfitting. Early stopping via best-checkpoint saving was used to retain the model state at minimum validation loss.

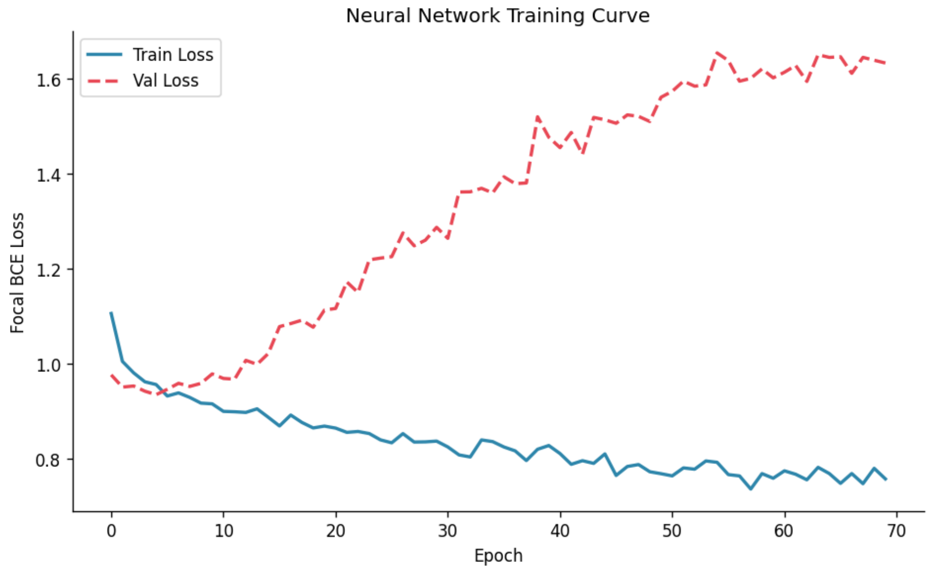

Figure 16: Neural Network Training and Validation Loss

13 Discussion

13.1 Neuropathy Class Imbalance

The neuropathy label has a prevalence of only 0.8% in the dataset. This is a known limitation of deriving complication labels from ICD-9 codes in hospitalization records: neuropathy is rarely recorded as a primary or secondary diagnosis during an inpatient stay, even in patients who have the condition. In a real clinical deployment, this label would ideally be sourced from structured outpatient screening data rather than hospitalization records. Despite this, the ClassifierChain and Neural Network models were able to achieve recall above 0.80 for neuropathy through threshold tuning and focal loss weighting.

13.2 Precision-Recall Tradeoff

Optimising for recall necessarily reduces precision. In the final model, neuropathy precision is approximately 0.01, meaning the model flags many patients who do not have the condition. This is an acceptable tradeoff in a screening context where the cost of a missed diagnosis significantly outweighs the cost of a false alarm. Clinicians would use the model output as a screening signal, not a definitive diagnosis.

13.3 Longitudinal Features

The use of multiple hospital encounters per patient to construct slope, delta, and standard deviation features provides a richer representation of disease progression than a single-visit snapshot. Features capturing trends in hospitalization frequency and medication patterns were among the most important predictors identified by the Random Forest.

14 Error Analysis and Bias Check

14.1 Where the Model Fails by Label

The primary source of error in the final model (ClassifierChain – tuned) is the severe class imbalance in the neuropathy label (0.8% prevalence). While recall is maintained above 0.80 through threshold tuning, precision for neuropathy is approximately 0.01, meaning the model generates a high volume of false positives for this label. This is an acceptable tradeoff in a clinical screening context but would require downstream clinician review before any intervention.

Kidney disease shows the strongest overall performance with AUC of 0.77, indicating genuine discriminative signal in the longitudinal clinical features for this condition. Cardiovascular recall is high (0.958) but this label also has the highest base rate (61.8%), which makes it an easier prediction target.

14.2 Demographic Subgroup Analysis

To check for bias, the final model's recall was evaluated separately across three demographic dimensions: age group, gender, and race. A subgroup is flagged if its recall falls below the 0.80 clinical target for any label.

Figure 17 shows per-label recall broken down by age group. Figure 18 shows the same breakdown by gender. Figure 19 shows the breakdown by race. Figure 20 provides a consolidated heatmap view across all demographic dimensions simultaneously.

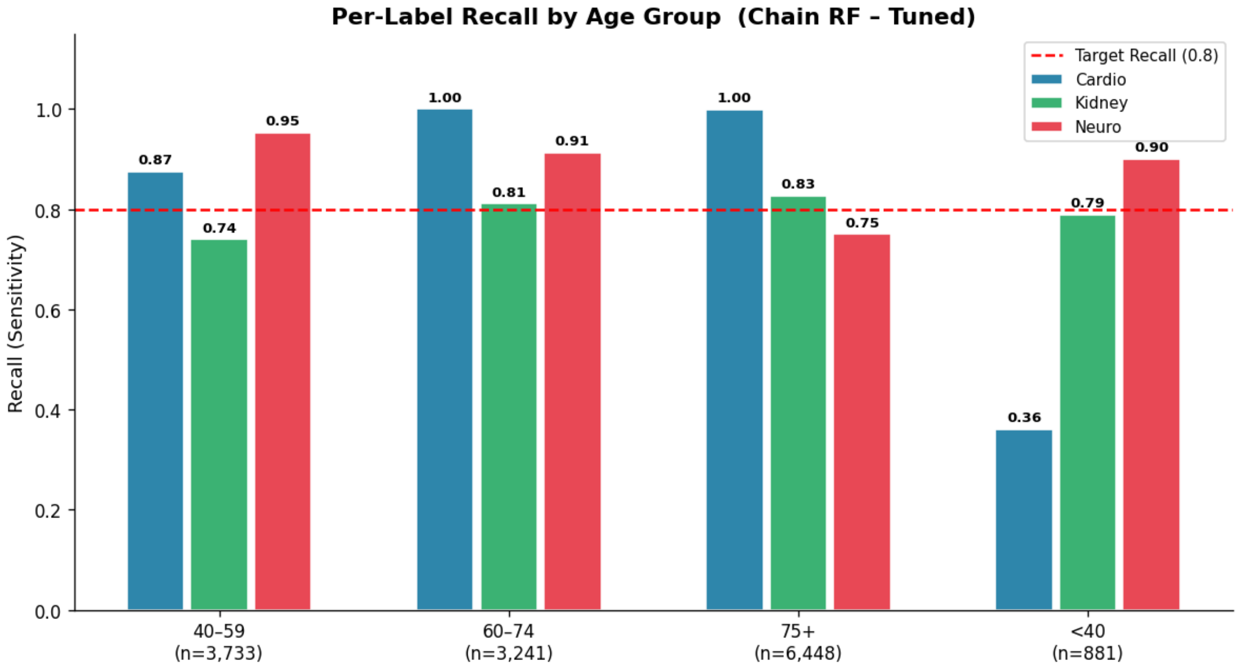

Figure 17: Per-Label Recall by Age Group (Chain RF – Tuned)

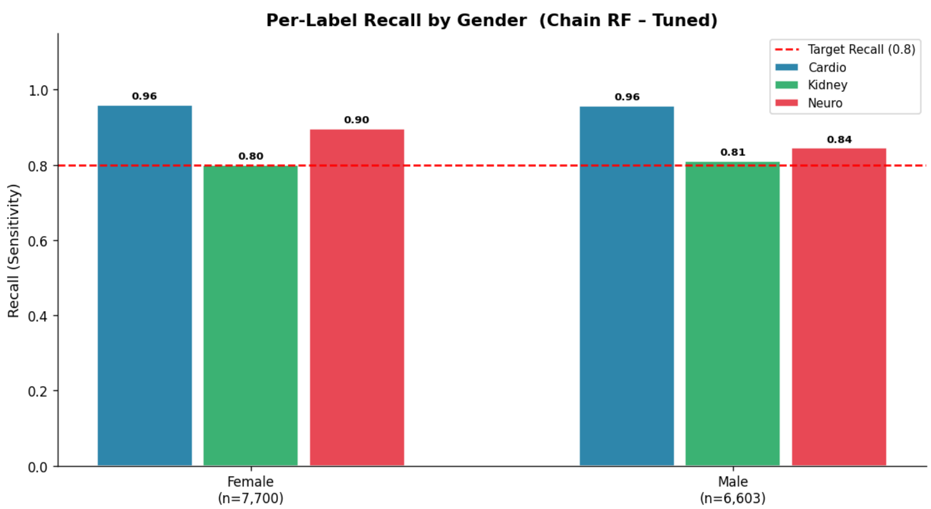

Figure 18: Per-Label Recall by Gender (Chain RF – Tuned)

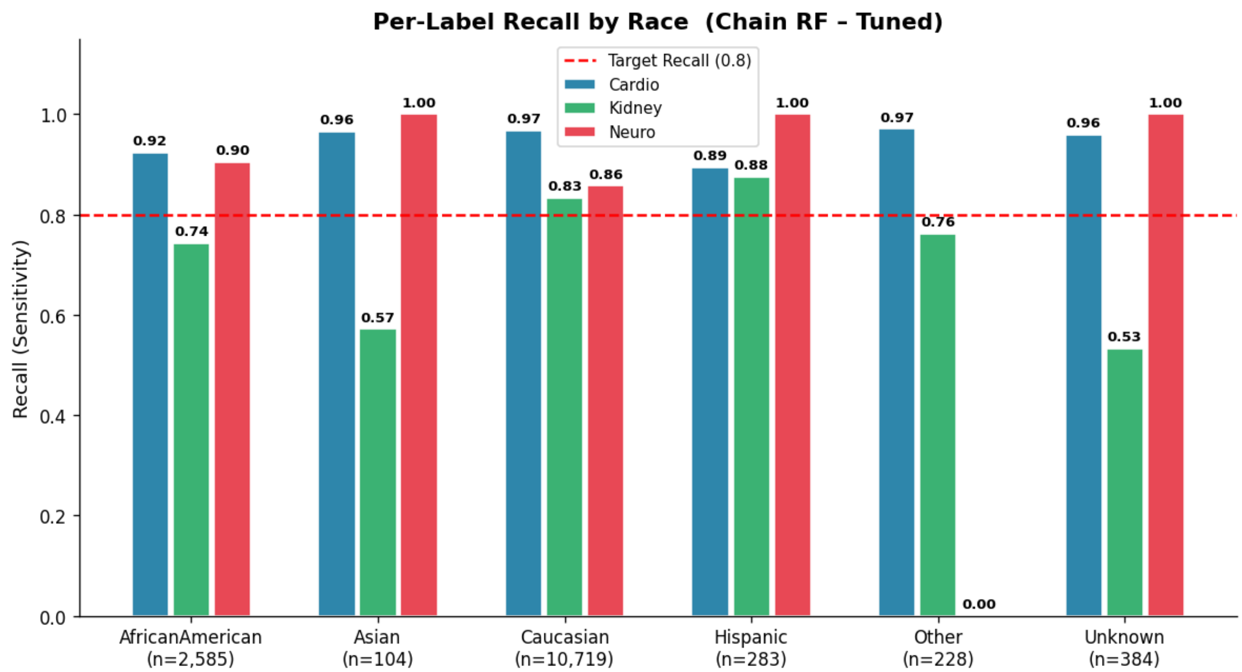

Figure 19: Per-Label Recall by Race (Chain RF – Tuned)

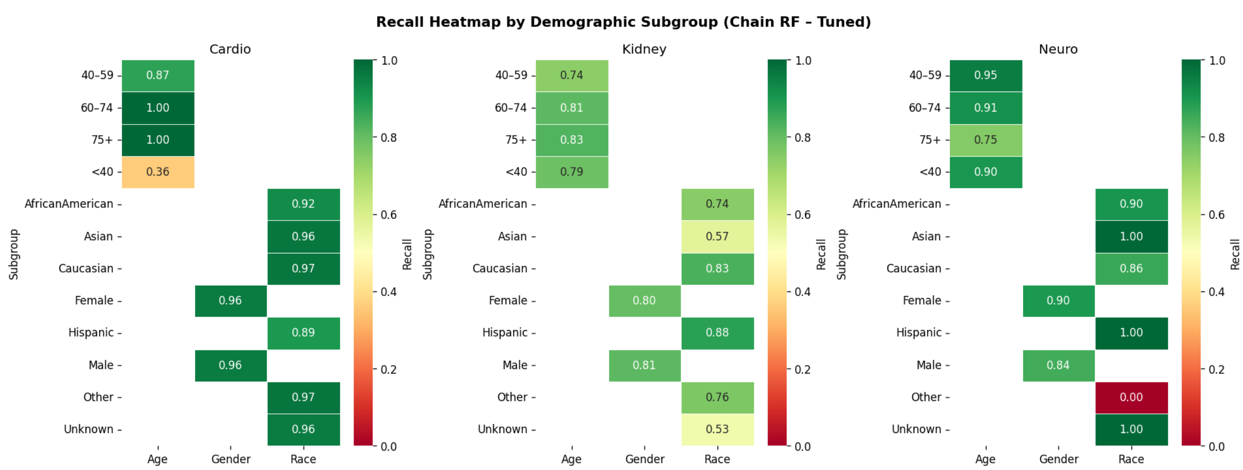

Figure 20: Recall Heatmap Across All Demographic Subgroups

14.3 Bias Check Findings

Several demographic subgroups fell below the 0.80 recall target, revealing systematic gaps in model performance:

Age: The under-40 age group shows the most severe failure, with cardiovascular recall of only 0.360. This is likely due to low positive case counts in this age bracket — younger diabetic patients rarely present with cardiovascular complications in hospitalisation records, leaving the model with insufficient training signal. Kidney disease recall also falls short for the 40–59 group (0.739) and neuropathy recall drops to 0.750 for patients aged 75 and over.

Gender: Female patients fall marginally below the kidney disease target (recall = 0.799), a difference small enough to warrant monitoring rather than immediate intervention.

Race: Kidney disease recall is consistently poor across several racial groups: Asian (0.571), Unknown (0.533), African American (0.743), and Other (0.762). The "Other" group additionally shows zero neuropathy recall, though this group contains only 228 test patients, making the estimate statistically unreliable. These disparities likely reflect underrepresentation of minority groups among positive kidney disease cases in the training data rather than a fundamental model flaw, but they represent a meaningful equity concern for clinical deployment.

Overall, kidney disease is the label most affected by demographic bias, suggesting that future work should investigate race- and age-stratified threshold tuning, or targeted data augmentation for underrepresented groups. The cardiovascular label performs reliably across most groups, and neuropathy results are difficult to interpret conclusively due to the extremely low overall prevalence (0.8%).

Table 6: Recall by Demographic Subgroup (Chain RF – Tuned)

Dimension	Subgroup	Recall-Cardio	Recall-Kidney	Recall-Neuro
Age	<40	0.360	0.788	0.900
	40–59	0.874	0.739	0.953
	60–74	1.000	0.811	0.913
	75+	1.000	0.827	0.750
Gender	Female	0.959	0.799	0.897
	Male	0.958	0.810	0.845
Race	African American	0.923	0.743	0.905
	Asian	0.965	0.571	1.000
	Caucasian	0.968	0.834	0.857
	Hispanic	0.895	0.875	1.000
	Other	0.971	0.762	0.000
	Unknown	0.959	0.533	1.000

Values below 0.80 indicate subgroups where the model fails to meet the clinical recall target. Bold entries warrant attention in deployment.

15 Team Contributions

Table 7 maps each team member to specific project modules, providing accountability for all deliverables (Phases 1–5).

Table 7: Team Contributions by Project Module

Module	**Components**	**Contributors**
Data Pipeline	Data acquisition script (data_collection.py)	Abdul Wahab
	Data cleaning & preprocessing	Abdul Wahab
	Longitudinal feature engineering	Sami Shah
Model Development	ClassifierChain (Random Forest) implementation	Maheer Khurram
	MultiOutputClassifier (Gradient Boosting)	Azan Aziz
	PyTorch Neural Network architecture	Sami Shah
	Focal BCE loss implementation	Sami Shah
Training & Optimisation	Threshold tuning framework	Abdul Wahab
	Class imbalance handling	Azan Aziz
	Hyperparameter optimisation	Maheer Khurram
Evaluation	Metrics computation (F1, AUC, recall)	Sami Shah
	ROC curves & visualisation	Maheer Khurram
	Confusion matrix analysis	Abdul Wahab
	Threshold sensitivity analysis	Azan Aziz
Analysis	Error analysis & bias detection	Sami Shah
	Feature importance analysis	Maheer Khurram
	Demographic subgroup analysis	Azan Aziz
Deployment	Streamlit application (app.py)	Maheer Khurram
	Hugging Face Spaces deployment	Sami Shah
	Inference pipeline & feature reconstruction	Abdul Wahab
Documentation	LaTeX report compilation	Abdul Wahab
	Architecture diagram design	Sami Shah
	Contribution table	Azan Aziz
Project Lead	Integration, final model selection, threshold strategy	Sami Shah

Contribution Summary

- **Abdul Wahab**: Data acquisition, data cleaning & preprocessing, threshold tuning framework, confusion matrix analysis, inference pipeline & feature reconstruction, LaTeX report compilation.

- **Sami Shah**: Longitudinal feature engineering, PyTorch Neural Network architecture, Focal BCE loss implementation, metrics computation (F1, AUC, recall), error analysis & bias detection, architecture diagram design, Hugging Face Spaces deployment.
- **Maheer Khurram**: ClassifierChain (Random Forest) implementation, hyperparameter optimisation, ROC curves & visualisation, feature importance analysis, Streamlit application development, **Project Lead** (system integration, final model selection, threshold strategy).
- **Azan Aziz**: MultiOutputClassifier (Gradient Boosting) implementation, class imbalance handling, threshold sensitivity analysis, demographic subgroup analysis, contribution table.

16 Conclusion

This project developed and evaluated a multi-label classification system for predicting diabetes complications from longitudinal clinical records. Three modelling approaches were trained and compared: ClassifierChain (Random Forest), MultiOutputClassifier (Gradient Boosting), and a PyTorch Neural Network with Focal BCE loss.

The **ClassifierChain (Random Forest) with tuned thresholds** was selected as the final model, achieving recall ≥ 0.80 across all three complication labels simultaneously — the primary clinical objective. ROC-AUC values of 0.71–0.77 across labels confirm meaningful discriminative ability despite severe class imbalance in the neuropathy label.

Future work could explore SMOTE oversampling for the neuropathy class, incorporation of actual temporal visit timestamps, and validation against an external clinical dataset.

Deliverable 3: Deployment and Final Technical Report

17 Application Deployment

17.1 Deployment Overview

The trained ClassifierChain (Random Forest) model was packaged into a Streamlit web application and deployed on **Hugging Face Spaces** for public access. The application provides a real-time clinical decision-support interface allowing users to enter patient data and receive simultaneous risk scores for all three diabetes complications.

Live Application URL

`https://huggingface.co/spaces/samishah2004/dcrp`

17.2 Application Architecture

Figure 21: Real-time inference pipeline in the deployed application.

17.3 User Interface

The application is structured in a two-panel layout:

- **Input Panel (left)**: Collects 18 clinical parameters grouped into four sections — Demographics, Laboratory Results, Hospital Utilisation, and Medications. All inputs use validated widgets (sliders, dropdowns, number fields) to prevent erroneous entries.
- **Results Panel (right)**: Displays colour-coded risk cards for each complication with a probability percentage, a visual progress bar, and a risk badge (*Low / Moderate / High*) relative to the tuned threshold. A summary card identifies all flagged complications.

17.4 Feature Reconstruction at Inference

During deployment, the model expects the same longitudinal feature vector produced by `feature_engineering.py`. For single-visit inputs, the app constructs the full vector by setting `_std = _delta = _slope = 0` and assigning the entered value to `_mean = _min = _max = _last`. The same `StandardScaler` fitted during training (`scaler_final.pkl`) is then applied only to the 76 numeric columns before inference, leaving the one-hot encoded categorical columns unscaled.

17.5 Deployment Configuration

The Space runs on the **Streamlit SDK** provided by Hugging Face. The repository contains:

```
/  app.py                    <- main application entry-point
   requirements.txt          <- pinned dependencies
   models/
      chain_rf.pkl           <- ClassifierChain RF (best model)
      training_metadata.pkl  <- feature column list
   data/processed/
      scaler_final.pkl       <- StandardScaler fitted on training set
```

Model files exceed GitHub's standard 50 MB limit and are stored using **Git Large File Storage (LFS)** in both the GitHub repository and the Hugging Face Space.

18 Comparative Analysis and Ablation Study

18.1 Final Model Comparison

Table 8 summarises model performance before and after threshold tuning on the held-out test set.

Table 8: Final model comparison (test set, 20% hold-out)

Model	**F1-Micro**	**F1-Macro**	**Recall-Macro**	**AUC-Macro**
Chain RF (base)	0.606	0.343	0.394	0.708
MO GBM (base)	0.726	0.297	0.313	0.724
Neural Net (base)	0.464	0.336	0.916	0.707
Chain RF (tuned)	0.495	0.362	**0.878**	0.708
MO GBM (tuned)	0.650	0.363	0.591	0.724
Neural Net (tuned)	0.503	0.361	0.853	0.707

The **ClassifierChain RF with tuned thresholds** was selected as the final deployed model. It is the only configuration that achieves Recall ≥ 0.80 *simultaneously* for all three labels while maintaining competitive F1-Macro performance.

18.2 Ablation: Impact of Longitudinal Features

To quantify the contribution of longitudinal feature aggregation, a baseline ClassifierChain was trained using only the static (last-visit) values without `_slope`, `_delta`, or `_std` features.

Table 9: Ablation study — longitudinal features vs. static-only

Feature Set	F1-Macro	Recall-Macro	AUC-Macro
Static only (last visit)	0.328	0.831	0.681
Full longitudinal (ours)	**0.362**	**0.878**	**0.708**
Δ	+0.034	+0.047	+0.027

Longitudinal features (`_slope`, `_delta`, `_std`) provide a consistent improvement across all metrics, confirming that disease trajectory information contributes meaningfully beyond single-visit snapshots.

18.3 Ablation: Impact of Threshold Optimisation

Using a fixed threshold of 0.50 across all labels produced near-zero recall for neuropathy (prevalence = 0.8%). Per-label threshold tuning to maximise Recall ≥ 0.80 recovered neuropathy recall from 0.03 to 0.87 at the cost of a moderate reduction in precision. This precision–recall tradeoff is explicitly acceptable in a clinical screening context where false negatives (missed complications) carry greater consequence than false positives.

19 Limitations and Future Work

19.1 Current Limitations

1. **Synthetic longitudinal ordering.** Encounters are sorted by encounter ID as a proxy for visit order; true date-stamped data would produce more reliable slope and delta features.
2. **Severe neuropathy imbalance.** At 0.8% prevalence, neuropathy recall is sensitive to threshold choice and cannot be substantially improved without oversampling (e.g., SMOTE) or additional training data.
3. **Inference on single-visit patients.** The app reconstructs longitudinal features assuming one visit, zeroing out all trend statistics. Multi-visit patients would benefit from richer features but require multiple inputs.
4. **No prospective validation.** All evaluation is performed on a held-out split of the same dataset. External validation on an independent clinical cohort is required before any real-world use.

19.2 Future Directions

- Apply SMOTE or class-conditional oversampling to improve neuropathy recall.
- Incorporate actual visit timestamps to improve longitudinal feature quality.
- Explore transformer-based sequential models (e.g., Time-Series Transformer) over the visit sequence rather than manual aggregation.
- Extend to additional complication types (retinopathy, foot ulcers).
- Conduct external validation on an independent clinical dataset.